

Technology Stack

Date	Team ID	Project Name
30 September 2026	SWTID-2026-9981	PocketSmart AI: Your Smart Budget & Recommendation Assistant

Table 1: Components and Technologies

S.No	Component	Description	Technology
1	User Interface	Web interface for sign-up, expense entry, dashboard, budgets, recommendations and chat	Streamlit (Python), Plotly charts
2	Application Logic - Core	Handles authentication, expense management, budget rules and alerts	Python 3, pandas
3	Application Logic - Categorisation	Predicts expense category from the description text	scikit-learn (TF-IDF with Logistic Regression)
4	Application Logic - AI Recommendations	Creates saving tips, product alternatives and chat answers from the spending summary	Google Gemini API (LLM) with prompt engineering
5	Database	Stores users, budgets and expenses	SQLite
6	File Storage	Product and alternative catalogue; CSV import and export	CSV files
7	External Service	Large language model for natural-language responses	Gemini API
8	Infrastructure	Public hosting of the application	Streamlit Community Cloud
9	Version Control	Source code and phase documents	Git and GitHub

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used to build the application	Streamlit, pandas, scikit-learn, Plotly
2	Security Implementation	Hashed passwords, per-user data access, API key stored as a secret	hashlib (SHA-256 with salt), Streamlit secrets
3	Scalable Architecture	Layered design with separate UI, logic and data modules	Modular Python packages; database can move to PostgreSQL
4	Availability	Always-on hosted application	Streamlit Community Cloud
5	Performance	Cached data loading and lightweight model for fast responses	Streamlit caching, joblib-saved model